Hardness of approximating deterministic communication complexity of partial Boolean functions

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Editorial note. Faithful editable transcription of the official CCC2020 CC BY3.0 publication, article22. Independent partial-Boolean communication-minimization theme: complete Section5.1 source definitions19–31 and cited coloring-hardness20/21 premises, complete Section5.2 graph-to-partial-matrix construction and Theorem32 author proof, complete Section5.3 original Corollaries33/34 with author deductions and Theorem35 full approximate-characterization impossibility proof. All82 official bibliography entries retained using the shared frozen formal-source bibliography. Original partial{0,1,*} matrix, unrestricted deterministic two-party protocol/leaf/depth/cover/DAG measures, approximation ratios/error and eventual monotonicity/gap conditions retained. No fixed-round or total-function theorem substituted. Original printed references/wording retained without mathematical repair, including Section5.3 reference to balanced protocol as proofTheorem20 and source DAG counting statement. Standard external graph-coloring hardness retained at author citation boundary, without recursive proof. All direct parameter consequences/communication measures grouped once. Finite comparison to NT065: that item uses CCC2021.31 deterministic totalBoolean fixed-round protocol-size hardness with a Boolean construction and round elimination; this item uses CCC2020.22 partialBoolean matrices, arbitrary depth and coloring-cover characterization. Distinct model, main result and mechanism. Native not available in bounded author/source lookup; all selected TeX transcribed from formally licensed PDF. No necessary figures. Root handles compilation/database/publication.

Allowed standard prerequisites (editorial).

- Gap hardness of graph coloring. Exact original Theorem20[82] and Theorem21[79] statements retained; no recursion into prior hardness proofs.

Theorem 35 (Author Theorem35)**.** *Let* $\lambda, \eta : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ *be functions such that* $\eta(1 + \log x) < \lambda(0.99x)$ *for all sufficiently large* x*. Moreover, assume that these functions are non-decreasing for every large enough* x*. Then, if* $\mathrm{P} \neq \mathrm{NP}$*, there is no polynomial-time computable function* r*, which accepts as input the communication matrix* $M_f \in \{0, 1, *\}^{n \times n}$ *of a partial Boolean function*

$f : [n] \times [n] \to \{0, 1, *\}$, *and for every large enough* n *outputs a value* $r(M_f)$ *such that*

$$\lambda(D(f)) \leq r(M_f) \leq \eta(D(f)).$$

5 On the NP-hardness of communication minimization problems

5.1 Background

Hardness of graph coloring Our NP-hardness reduction is from the chromatic number problem:

Definition 19 (Author Definition19: Chromatic number)**.** *A coloring of an undirected graph* G*, is a partition of the vertices such that no edge has both endpoints in the same part. The chromatic number of a graph* G*, denoted* $\chi(G)$*, is the smallest number of parts of a coloring of* G*.*

The NP-hardness of approximating the chromatic number has been established by a series of results [57, 31, 24], culminating in a paper by Zuckerman [82], where the following was proven:

Theorem 20 (Author Theorem20, [82])**.** *For every constant* $\epsilon > 0$ *it is NP-hard to approximate* $\chi(G)$ *for a given* n*-vertex graph* G*, with an approximation ratio better than* $n^{1-\epsilon}$*. More precisely, for every* $L \in \text{NP}$ *and* $\epsilon > 0$*, there exists a polynomial-time algorithm that, on input* x*, outputs a parameter* k *and an* n*-vertex graph* G *such that if* $x \in L$ *then* $\chi(G) \leq k$*, and if* $x \notin L$ *then* $\chi(G) > n^{1-\epsilon} \cdot k$*.*

In the reductions above, the parameter k is $\Theta(n^\epsilon)$. More recent results on the hardness of approximating chromatic number allow for a different gap, where k is constant as n grows, and we wish to distinguish graphs which are k-colorable from graphs which are not $g(k)$-colorable, for a fast-growing function $g : \mathbb{N} \to \mathbb{N}$. An original such result with $g(k) = k^{\Omega(\log k)}$ was shown by Khot [47], which was later improved to $g(k) = 2^{\Omega(k^{1/3})}$ by Huang [37]. The latest result, by Wrochna and Živný [79], achieves $g(k) = \binom{k}{\lfloor k/2 \rfloor}$:

Theorem 21 (Author Theorem21, [79])**.** *Let* $k \geq 4$ *be a natural number. For every* $L \in \text{NP}$*, there exists a polynomial-time algorithm that, on input* x*, outputs an graph* G *such that if* $x \in L$ *then* $\chi(G) \leq k$*, and if* $x \notin L$ *then* $\chi(G) > \binom{k}{\lfloor k/2 \rfloor}$*.*

Communication complexity of relations and partial functions We will need the following definitions.

Definition 22 (Author Definition22)**.** *We will call a two-player communication relation, or simply a relation, to any subset* $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$*, where* $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ *are finite sets, such that, for every* $(x, y) \in \mathcal{X} \times \mathcal{Y}$*, there exists at least one* $z \in \mathcal{Z}$ *with* $(x, y, z) \in F$*.*

Given a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, and a pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$, we let

$$F(x, y) = \{z \in \mathcal{Z} \mid (x, y, z) \in F\} \neq \varnothing.$$

The matrix with $\mathcal{X}$-indexed rows and $\mathcal{Y}$-indexed columns, whose (x, y) entry is $F(x, y)$, is called the communication matrix of F. Given a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, a rectangle $R = A \times B \subseteq \mathcal{X} \times \mathcal{Y}$, and an element $z \in \mathcal{Z}$, we say R is z-monochromatic for F, if $(x, y, z) \in F$ for all $(x, y) \in R$. We say R is monochromatic for F if there exists a $z \in \mathcal{Z}$ with R being z-monochromatic for F. A partial two-player function is a relation $f \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ such that, for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$, either $|f(x, y)| = 1$, in which case we say f is defined at (x, y), or $f(x, y) = \mathcal{Z}$, in which case we say f is undefined at (x, y). If f is defined at (x, y), we write $f(x, y) = z$ in place of $f(x, y) = \{z\}$, and if f is undefined at (x, y), we write $f(x, y) = *$, instead of $f(x, y) = \mathcal{Z}$.

Definition 23 (Author Definition23: $P(F)$)**.** *Given a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, a partition of F is a family Π of monochromatic rectangles for F, such that every $(x, y) \in \mathcal{X} \times \mathcal{Y}$ appears in exactly one rectangle of Π. The partition number of a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, denoted $P(F)$ is the smallest possible size of a partition of F.*

Definition 24 (Author Definition24: $C(F)$)**.** *Given a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, a cover of F is a family Γ of monochromatic rectangles for F, such that every $(x, y) \in \mathcal{X} \times \mathcal{Y}$ appears in at least one rectangle of Γ. The cover number of a relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, denoted $C(F)$ is the smallest possible size of a cover of F.*

In the context of Boolean-valued functions, the usual notion of cover number (see [42] Section4.2,p97), and related notion of non-deterministic communication complexity, only require that the 1s of the communication matrix are covered.

Definition 25 (Author Definition25: $C_1(F)$)**.** *Given a partial two-player function $f : \mathcal{X} \times \mathcal{Y} \to \{0, 1\}$, a 1-cover of F is a family Γ of 1-monochromatic rectangles for F, such that every $(x, y) \in \mathcal{X} \times \mathcal{Y}$ having $f(x, y) = 1$ appears in at least one rectangle of Γ. The1-cover number of f, denoted $C_1(F)$ is the smallest possible size of a 1-cover of F. We then define the non-deterministic communication complexity of f to be $N(f) = \lceil \log C_1(f) \rceil$.*

Definition 26 (Author Definition26)**.** *A protocol π over $\mathcal{X} \times \mathcal{Y}$ is a rooted tree:*

- *Each node v is associated with a rectangle $\pi^{-1}(v) = A \times B \subseteq \mathcal{X} \times \mathcal{Y}$.*
- *Each non-leaf node v, with $\pi^{-1}(v) = A \times B$, is labeled by either(a) a partition $A = A_0 \dot\cup A_1$ of A, in which case we say it is Alice's node or(b) a partition $B = B_0 \dot\cup B_1$ of B, in which case we say it is Bob's node.*
- *The rectangle associated with the root is $\mathcal{X} \times \mathcal{Y}$.*
- *If a non-leaf node v of Alice has $\pi^{-1}(v) = A \times B$ and is labeled by a partition $A = A_0 \dot\cup A_1$ of A, then for each $c \in \{0, 1\}$ there will be a unique child v_c of v, with $\pi^{-1}(v_c) = A_c \times B$; similarly for Bob's nodes.*

Definition 27 (Author Definition27: $L(F), D(F)$)**.** *Let $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ be a relation, and π a protocol over $\mathcal{X} \times \mathcal{Y}$. We say that π is a protocol for F if, for any leaf ℓ of π, the rectangle $\pi^{-1}(\ell)$ is monochromatic for F. We then let $L(F)$ be the smallest possible number of leaves in a protocol for F, and we let $D(F)$ be the smallest possible depth of a protocol for F.*

Note that the rectangles associated with the leaves of a protocol for F form a partition of F, any partition of F is a cover of F, and any cover of F contains a 1-cover of F, and hence we get the following:

Lemma 28 (Author Lemma28)**.** *For any relation $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, $L(F) \geq P(F) \geq C(F)$, and for a partial two-player function $f : \mathcal{X} \times \mathcal{Y} \to \{0,1\}$, $C(f) \geq C_1(f)$.*

Definition 29 (Author Definition29)**.** *A DAG-like protocol γ over $\mathcal{X} \times \mathcal{Y}$ is a directed acyclic graph:*

- *γ has a single source node, called the root, and one or more sink nodes, called leaves.*
- *Each node v is associated with a rectangle $\gamma^{-1}(v) = A \times B \subseteq \mathcal{X} \times \mathcal{Y}$.*
- *Each non-leaf node v, with $\gamma^{-1}(v) = A \times B$, is labeled by either(a) a partition $A = A_0 \dot\cup A_1$ of A, in which case we say it is Alice's node or(b) a partition $B = B_0 \dot\cup B_1$ of B, in which case we say it is Bob's node.*
- *The rectangle associated with the root is $\mathcal{X} \times \mathcal{Y}$.*
- *If a non-leaf node v of Alice has $\gamma^{-1}(v) = A \times B$ and is labeled by a partition $A = A_0 \dot\cup A_1$ of A, then for each $c \in \{0,1\}$ there will be an edge from v to a node v_c of γ, which must be such that $A_c \times B \subseteq \gamma^{-1}(v_c)$; similarly for Bob's nodes.*

Definition 30 (Author Definition30: $S(F)$)**.** *Let $F \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ be a relation, and γ a DAG-like protocol over $\mathcal{X} \times \mathcal{Y}$. We say that γ is a DAG-like protocol for F if, for any leaf ℓ of γ, the rectangle $\gamma^{-1}(\ell)$ is monochromatic for F. We then let $S(F)$ be the smallest possible number of nodes in a DAG-like protocol for F.*

Note that:(1) a protocol with k leaves is a DAG-like protocol with $2k-1$ nodes,(2) the rectangles associated with the leaves of a DAG-like protocol form a cover of F, and(3) any DAG with 1 source node,k sink nodes, and maximum out-degree 2 has at least $2k-1$ nodes in total. Hence:

Lemma 31 (Author Lemma31)**.** *$S(F) \leq 2L(F) - 1$ and $S(F) \geq 2C(F) - 1$.*

5.2 A reduction from Graph Coloring to Partial-MCCP

Let us be given a graph $G = ([n], E)$ with $E \subseteq \binom{[n]}{2}$, and consider the partial two-player Boolean function $f_G : [n] \times [n] \to \{0, 1, *\}$, given by

$$f_G(i,j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } \{i,j\} \in E, \\ * & \text{if } \{i,j\} \notin E. \end{cases}$$

By $f_G(i,j) = *$ we mean that $f_G(i,j)$ is undefined, which is to say Alice and Bob may output any value when given (i,j) as input.

Theorem 32 (Author Theorem32)**.** *We have $L(f_G) \leq 2\chi(G)$ and $C_1(f_G) = \chi(G)$.*

Proof. To show that $L(f_G) \leq 2\chi(G)$, consider the simple protocol where Alice and Bob have agreed on a coloring of G, so Alice begins by sending the color of her vertex, and Bob replies whether the color of his vertex is the same. If the two colors match, they output 1, and otherwise they output 0. This is a protocol for f, since a valid coloring of G will only color i and j by the same color if $i = j$, or $\{i, j\} \notin E$, in which case 1 is a valid output. If i and j are colored differently, then $i \neq j$, and so 0 is a valid output. This protocol has $2\chi(G)$ leaves. To show that $C_1(f_G) \leq \chi(G)$, we cover the diagonal by 1-monochromatic rectangles; each such rectangle corresponds to a color c, and is the smallest rectangle containing all diagonal entries (i, i) for vertices i of G that are colored c. Now suppose we have any positive cover Γ for f. The diagonal entries must be covered by 1-monochromatic rectangles of Γ, so consider the coloring of G which colors vertex $i \in [n]$ by the 1-monochromatic rectangle which covers (i, i). By our choice of f, any two diagonal entries belonging to the same 1-monochromatic rectangle are not connected by an edge of G. And so this gives us a valid coloring of G, which implies that $\chi(G) \leq |\Gamma|$. □

5.3 Consequences of the reduction

We may now take Theorems32 and20, together with Lemmas28 and31, to obtain:

Corollary 33 (Author Corollary33)**.** *For any constant $\epsilon > 0$, it is NP-hard to approximate $L(f), P(f), C(f), C_1(f)$ and $S(f)$, for a given partial function $f : [n] \times [n] \to \{0, 1, *\}$, with an approximation ratio better than $n^{1-\epsilon}$.*

Observe that this hardness of approximation result with a ratio of $n^{1-\epsilon}$ is very close to optimal, since all aforementioned measures for a partial two-player function $f : [n] \times [n] \to \{0, 1, *\}$ are upper-bounded by $2n$. Improving upon this might still be possible, and we leave it as an open problem. Now notice that the protocol given in the proof of Theorem20 is balanced. For this reason, its depth is logarithmic in the number of leaves. Thus, since computing $L(f)$ is hard up to an approximation ratio of $n^{1-\epsilon}$, it follows that computing the communication complexity $D(f)$ for a given partial function $f : [n] \times [n] \to \{0, 1, *\}$ is hard, even if we allow for an additive error term of $(1 - \epsilon) \log n$. Since $D(f) \leq \log n$, this implies that we cannot approximate $D(f)$ with an approximation ratio better than $\approx 1/\epsilon$, which we can take as an arbitrarily large constant. The same reasoning applies to non-deterministic communication complexity.

Corollary 34 (Author Corollary34)**.** *For any constant $\epsilon > 0$, it is NP-hard to approximate $D(f)$ or $N(f)$, for a given partial function $f : [n] \times [n] \to \{0, 1, *\}$, with an error term smaller than $(1 - \epsilon) \log n$. For any constant $c > 1$, it is NP-hard to approximate $D(f)$ or $N(f)$, for a given partial function $f : [n] \times [n] \to \{0, 1, *\}$, with an approximation ratio better than c.*

Let us now prove that, assuming P $\neq$ NP, there is no polynomial-time computable approximate characterization of the communication complexity of a partial Boolean function. This is captured by a more general result (Theorem35, stated at the start of this item), which is an immediate consequence of the reduction presented in Section5.2 and Theorem21.

Author proof of Theorem35. Suppose that functions λ, η, and r exist as described in the statement of the theorem. Let L be any language in NP, and

assume that k is a sufficiently large constant. For any $x \in \{0,1\}^n$, let G_x be the corresponding graph given by Theorem21. Then let $f_x = f_{G_x}$ be as defined in Section5.2 above, and let M_x be the communication matrix of f_x. If $x \in L$, we then have by Theorem32 that $k \geq \chi(G_x) \geq \frac{1}{2}L(f_x)$, and because the protocol for f_x that witnesses this fact is balanced, we get $1 + \log k \geq D(f_x)$. Hence $\eta(1 + \log k) \geq \eta(D(f_x)) \geq r(M_x)$. If, on the other hand,$y \notin L$, then we have $2^{0.99k} < \binom{k}{\lfloor k/2 \rfloor} < \chi(G_y) = C_1(f_y) \leq L(f_y)$, and thus $0.99k < D(f_y)$. Then $\lambda(0.99k) \leq \lambda(D(f_y)) \leq r(M_y)$. But since $\lambda(0.99k) > \eta(1 + \log k)$ using our assumptions on these functions and on k, it follows that $r(M_y) > r(M_x)$. As a consequence, the polynomial time function r can be used to distinguish positive and negative instances of L. Since L is an arbitrary language in NP, we get that P = NP. This completes the proof. □

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